E. Flag 2

time limit per test: 2 seconds
memory limit per test: 128 megabytes

According to a new ISO standard, a flag of every country should have, strangely enough, a chequered field $n \times m$, each square should be wholly painted one of 26 colours. The following restrictions are set:

- In each row at most two different colours can be used.
- No two adjacent squares can be painted the same colour.

Pay attention, please, that in one column more than two different colours can be used.

Berland's government took a decision to introduce changes into their country's flag in accordance with the new standard, at the same time they want these changes to be minimal. By the given description of Berland's flag you should find out the minimum amount of squares that need to be painted different colour to make the flag meet the new ISO standard. You are as well to build one of the possible variants of the new Berland's flag.

Input

The first input line contains 2 integers n and m $(1 \le n, m \le 500)$ — amount of rows and columns in Berland's flag respectively. Then there follows the flag's description: each of the following n lines contains m characters. Each character is a letter from `a` to `z`, and it stands for the colour of the corresponding square.

Output

In the first line output the minimum amount of squares that need to be repainted to make the flag meet the new ISO standard. The following n lines should contain one of the possible variants of the new flag. Don't forget that the variant of the flag, proposed by you, should be derived from the old flag with the minimum amount of repainted squares. If the answer isn't unique, output any.

Examples

input Copy

```
3 4
aaaa
bbbb
cccc
```

output Copy

```
6
abab
baba
acac
```

input Copy

```
3 3
aba
aba
zzz
```

output Copy

```
4
aba
bab
zbz
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 18 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371091750	Apr/14/2026 10:18	Accepted

→ Problem tags

dp *2000

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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